NATHANAEL JASON NABABAN

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Electrical Engineering & Computer Science Student | Embedded Systems, IoT, Applied AI

Third-year EECS student at National Tsing Hua University with hands-on experience in embedded systems, IoT, power electronics, energy monitoring, and applied machine learning. Experienced in sensor integration, MQTT communication, hardware testing, simulation, and translating prototypes into field applications. Seeking an applied R&D internship at ITRI.

SELECTED ENGINEERING PROJECTS

Two-Phase Interleaved Buck Converter | Power Electronics Research Project | 2025-2026

- Implemented 20 kHz PWM, LC filtering, and over-voltage/over-current protection; measured approximately 140 mA total current ripple near 50% duty cycle and identified closed-loop control and active current sharing as the next development steps.
- Designed and experimentally validated a DSP/FPGA-controlled two-phase interleaved buck converter; used PLECS simulation and hardware testing to verify CCM operation, proportional voltage conversion, and ripple cancellation across duty cycles of 0.25, 0.50, and 0.75.

Cultivator - Aerator Health Monitoring System | *Co-founder / Product & Business Development* | *2026*

- Built an aquaculture equipment-monitoring concept that combines electrical-current and vibration signatures to identify abnormal aerator behavior and provide simple equipment-health alerts.
- Coordinated embedded-system development, farmer validation, pilot planning, and partnership outreach across multiple Asian markets; helped secure pilot commitments, including paid pilots.
- Defined product requirements around field reliability, low-cost deployment, MQTT-based telemetry, fault-data collection, and future machine-learning classification.

Autonomous Embedded Systems Vehicle | *NTHU Embedded Systems Project* | *2026*

- Integrated STM32 boards, Pixy2 vision, LaserPING distance sensing, continuous-rotation servo feedback, MQTT, and an LVGL touchscreen interface.
- Implemented line following, barcode-based navigation, obstacle detection, remote commands, telemetry, and real-time path visualization using C/C++ on Mbed OS.
- Debugged networking, sensor communication, memory faults, control tuning, and multi-board system integration under time-constrained demonstration requirements.

IoT Solar-Energy Analytics Platform | *First Prize - Smart Energy Saving and IoT Application*

- Developed an NB-IoT and MQTT-based platform for real-time solar-energy monitoring and analytics.
- Connected sensing, communications, and data visualization to support energy-efficiency analysis and practical deployment.

Applied Machine Learning Projects | *Python / PyTorch / scikit-learn*

- Built and evaluated regression, decision-tree, random-forest, SVM, and neural-network models for structured and sensor-derived datasets.
- Performed preprocessing, model selection, hyperparameter comparison, and evaluation using accuracy, F1 score, and validation results.

TECHNICAL SKILLS

Embedded & IoT: C/C++, Python, STM32, Mbed OS, MQTT, NB-IoT, SPI, I2C, UART, sensor integration, LVGL
Data & Engineering: Machine learning, PyTorch, scikit-learn, MATLAB, PLECS, signal analysis, control-system debugging
Product & Collaboration: Technical documentation, field validation, market research, partnership development, project coordination
Languages: Indonesian (native), English (C1), Mandarin Chinese (B1)

EDUCATION

National Tsing Hua University | *B.S. Electrical Engineering and Computer Science* | *Hsinchu, Taiwan* | *Sep 2023-Present*

- GPA: 3.7/4.3; top 25% of class. Relevant study includes embedded systems, machine learning, probability and statistics, electrical machinery, signals and systems, and computer systems.

LEADERSHIP & EXPERIENCE

OCHISA Student Organization | *Media Team Leader* | *Aug 2024-Aug 2025*

- Led planning and documentation for major initiatives; used membership and social-media data to improve outreach and participation.

AWARDS

National Champion, Hult Prize Taiwan - Cultivator smart aquaculture monitoring startup
First Prize, Smart Energy Saving and IoT Application - IoT solar-energy analytics platform